

WTW 158 - **WORKSHEET ON DIFFERENTIATION -2013**

Find the derivatives of the following functions. Don't simplify your answers.

1. $f(x) = x^2 e^{4x}$
2. $f(\theta) = \frac{\tan 2\theta}{1 + \sec(\theta^2)}$ (Use the quotient rule)
3. $f(t) = t^3 + e^t + 3^t$
4. $f(t) = (1 - 4t)^3 + e^{1-5t} + 3^{1-6t}$
5. $f(x) = 5 \sin(x^4)$
6. $f(x) = \sin^4 5x$
7. $f(t) = e^{\cos 5t}$
8. $f(t) = \cos(e^{5t})$
9. $f(u) = \ln(1 - \sqrt[3]{u})$
10. $f(u) = \sqrt[3]{1 - \ln 2u}$
11. $f(x) = \sec(\ln(2x + 6))$
12. $f(t) = \ln(\sec(3x + 7))$
13. $y = \operatorname{cosec}(3x^2)$
14. $y = 3^{\operatorname{cosec} x^2}$
15. $f(x) = \tan^{-1}(e^{8x})$ (Notation: $\tan^{-1} = \arctan = \operatorname{bgtan}$)
16. $f(t) = \sin^{-1}(1 - 9t^4)$ (Notation: $\sin^{-1} = \arcsin = \operatorname{bgsin}$)
17. $x^3 \sin y + xy^4 = y + e$
18. $f(t) = t^{\cos 2t}$
19. Find the equation of the tangent line to the graph of $f(x) = 2e^{1-x}$ where $x = 1$.
20. Find all x values in the domain of $f(x) = \frac{\ln 2x}{x^3}$ where you can draw a tangent line parallel to the x -axis

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